

Middleware Architectures 2

Motivation and Course Overview

doc. Ing. Tomáš Vitvar, Ph.D.

tomas@vitvar.com • @TomasVitvar • <https://vitvar.com>



Czech Technical University in Prague

Faculty of Information Technologies • Software and Web Engineering • <https://vitvar.com/lectures>



Evropský sociální fond
Praha & EU: Investujeme do vaší budoucnosti

Modified: Mon Feb 16 2026, 11:51:47
Humla v1.0

Overview

- Motivation
- Scope, Requirements, Learnings
- Assessment, Resources, Communication

Web 2.0

- A new version of the Web?
- Principles
 - *Read-Write Web*
 - *Programmable Web*
 - *Realtime Web*
 - *Social Web*

Motivation in Brief

- Need for highly performant and scalable apps
 - *Sudden increase in traffic*
 - *Slashdot effect*
- The Web is programmable
 - *Applications provide **data** and **functionality***
 - *Users – end-users (GUI) and programmers (API)*
 - *Any company with a Web presence has an API*
 - *Google, Amazon, LinkedIn, Facebook, ...*
 - *Complex backends (microservices, API gateways, etc.)*

AM2 and AM1 Courses

- **AM2 builds on AM1**
- **Application Architecture**
 - *Multi-tier client-server architecture*
 - *Cloud native and microservices*
 - *Interface of the app, REST, gRPC*
 - *Client side of the architecture, JavaScript, AJAX*
 - *Infrastructure empowered by cloud technologies*
- **Technology, Platform**
 - *JavaScript*
 - *client-side + related technologies*
 - *server-side – asynchronous I/O, node.js*
 - *It does not mean you cannot combine technologies*
 - *Node.js as a Web server, ESB for proxy services with back-end systems, all running in a cloud environment (auto scaling, load balancers, message queues, etc.)*

Overview

- Motivation
- Scope, Requirements, Learnings
- Assessment, Resources, Communication

Scope

- Advanced HTTP
 - *Same origin policy, cross-origin*
 - *OAuth, Open ID, JWT*
 - *Realtime Web protocols*
- Cloud Architectures
 - *Details of some IaaS and PaaS services*
 - *Cloud Native, more microservice architecture*
 - *More Kubernetes*

Organization of Lectures

- 12 Lectures
 - Czech: *Mon 11:00-12:30, TH:A-s134*
- Plan
 - 16.02.2025 – *Motivation and Course Overview*
 - 23.02.2025 – *Asynchronous I/O*
 - 02.03.2025 – *Browser Networking (SOP, CORS)*
 - 09.03.2025 – *Security (concepts, mTLS, JWT)*
 - 16.03.2025 – *Security (OAuth, OpenID)*
 - 23.03.2025 – *Protocols for the Realtime Web (streaming, SSE)*
 - 30.03.2025 – *Protocols for the Realtime Web (WebSocket)*
 - 06.04.2025 – *Easters*
 - 13.04.2025 – *Cloud Architectures (introduction, IaaS)*
 - 20.04.2025 – *Kubernetes Architecture (overview, containers)*
 - 27.04.2025 – *Kubernetes Architecture (scheduling)*
 - 04.05.2025 – *Kubernetes Architecture (fault-tolerance)*
 - 11.05.2025 – *Kubernetes Performance (self-healing, scalability)*

Organization of Practicals

- Work alone, you can collaborate
- Practicals every second week
- Number of sessions: 6-7, 5 major tasks
 1. *Introduction, JavaScript*
 2. *CORS/JSONP*
 3. *OAuth (JWT)*
 4. *Realtime Web*
 5. *TBD*
- Plus a number of tasks to complete at home

Overview

- Motivation
- Scope, Requirements, Learnings
- **Assessment, Resources, Communication**

Assessment

- Details available in Courses:
<https://courses.fit.cvut.cz/NI-AM2/classification/index.html>

Assessment – Final Marks

Mark	Points	In words
A	100–90	výborně
B	89–80	velmi dobře
C	79–70	dobře
D	69–60	uspokojivě
E	59–50	dostatečně
F	49–0	nedostatečně

Source: <http://www.cvut.cz/pracoviste/pravni-odbor/dokumenty/studijni-predpisy/studijnirad.pdf>

- Everything good and bad will count
 - *practicals, coding, (pro-)activity, passiveness, hacking, lectures, exam, cheating, ...*

Resources

- Online sources
 - <https://courses.fit.cvut.cz/NI-AM2/> – lectures and labs information
 - <http://w20.vitvar.com> – both html and pdf (1 and 2 slides per page)
- Books
 - I. Grikorik: *High Performance Browser Networking*, O'Reily 2013
 - B. Burns: *Designing Distributed Systems*, O'Reily 2018
 - L. Richardson, M. Amundsen: *RESTful Web APIs*, O'Reilly Media, May 2015, ISBN 978-1-449-35806-8.
- Other
 - Many sources on the Web, to be listed throughout the course
 - A lot of W3C sources, Web architecture, HTTP

About Slides

- Humla – Open Source HTML5 Presentation Environment
 - *every slide has a unique URL*
 - *all figures linked with Google drawings*
 - *possible to format and print in PDF*
 - *running local, with back-end NodeJS support, and offline*
 - *Fork it at [Humla github repo](#)*
- Keys
 - 1 *default browsing mode*
 - 2 *slideshow mode (automatically scales to full screen)*
 - 3 *grid (overview) mode*
 - 4 *print mode, 2 slides per page*
 - ← *slide left*
 - *slide right*
 - d *debug mode*
 - e *toggle last error messages on/off*